

Initial Project Plan

LSTM vs. Transformer Encoder for Text Classification (AG News)

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1. Team

- Team: Jihun Jang (ETRI School, 02521122).
- Team leader: Jihun Jang (solo team).

2. Dataset Split

- HuggingFace `ag_news`, single source, 4 classes (labels 0–3, no conversion needed).
- `train_test_split(test_size=0.1, seed=42)` → train 108k / val 12k / test 7.6k. Classes balanced.
- The test set is used for final evaluation only (no model selection, tuning, or early stopping).

3. Preprocessing

- Tokenization: word level (lowercased, punctuation split).
- Vocabulary: built from train only, max 20k, minimum frequency 2, with PAD and UNK tokens.
- Maximum length 128 (truncate if longer, pad if shorter); padding is masked.
- The same input is used for both models.

4. Model Configuration

	LSTM	Transformer Encoder
Embedding	128	128 ($= d_{\text{model}}$)
Core	BiLSTM, 2 layers, hidden 128	2 layers, 4 attention heads, FFN 256, sinusoidal positional encoding
Pooling	masked mean	masked mean
Output / Loss	Linear → 4, Cross-Entropy	Linear → 4, Cross-Entropy

- Common: Adam, learning rate 1e-3, batch 64, up to 8 epochs, dropout 0.1 (common start, tuned by validation), seed 42, model selection by validation only.
- Parameter counts: LSTM $\approx$ 3.22M / Transformer $\approx$ 2.83M (comparable scale). No pre-trained models.

5. Ablation Plan

- Topic: data efficiency — train 5 / 25 / 50 / 100% ($\approx$ 5k / 27k / 54k / 108k).
- Hypothesis: with little data the Transformer is disadvantaged (weaker inductive bias) while the LSTM is stable on small data; the gap shrinks as data grows.
- Controls: the vocabulary is built once on the full training set and fixed/reused; subsets keep class proportions (stratified) with a fixed seed; val/test always use the full set.
- Figure: x = data size, y = accuracy / macro-F1, both models on one graph; the 100% point is reused as the main comparison.
- Optional: Transformer internal analysis — positional encoding on/off, attention-weight visualization, gradient saliency (input attribution).

6. Expected Analysis

- Performance: both around 90%+, with the Transformer slightly ahead or comparable.
- Data efficiency: at 5% LSTM $\geq$ Transformer, at 100% comparable (hypothesis test).
- Convergence: the Transformer is fast but learning-rate sensitive; the LSTM is stable (checked via loss curves).
- Confusion: Business $\leftrightarrow$ Sci/Tech expected to be largest. Analyze ≥ 5 misclassifications (shared / LSTM-only / Transformer-only): ambiguous wording, short sentences, truncation.
- Mechanism: accuracy maintained with positional encoding off would indicate AG News is vocabulary-centric (bag-of-words); attention and saliency should concentrate on class keywords.

Environment: Mac MPS / Colab T4, seed 42, checkpoints to Drive. Deliverables: Jupyter notebook + report (4–6 pp.) + AI Usage Appendix (≤ 1 p.).